

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT : ARTIFICIAL INTELLIGENCE

AIM: Breadth First Search & Iterative Depth First Search in Python.

- A) Implement the Breadth First Search algorithm to solve a given problem.
- B) Implement the Iterative Depth First Search algorithm to solve the same problem.
- C) Compare the performance and efficiency of both algorithms.

A) OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\itlab\Desktop\T076.py =====
Breadth First Search (BFS)
-----
Source      : Versova
Destination : Bandra
Shortest Path Found:
Versova -> Andheri -> Western Express Highway -> Bandra
Total Steps: 3
>>>
```

B) OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\itlab\Desktop\T076.py =====
Iterative Depth First Search (DFS)
-----
Source      : Versova
Destination : Bandra
Path Found:
Versova -> Lokhandwala -> Oshwara -> Andheri West -> Bandra
Total Steps: 4
>>>
```

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C) OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
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>>>
===== RESTART: C:\Users\itlab\Desktop\T076.py =====
===== BFS =====
Path : Versova -> Andheri -> Western Express Highway -> Bandra
Steps: 3
Nodes Visited: 8
Execution Time: 4.4099999286117963e-05 seconds
===== DFS =====
Path : Versova -> Lokhandwala -> Oshiwara -> Andheri West -> Bandra
Steps: 4
Nodes Visited: 5
Execution Time: 2.099999983329326e-05 seconds
===== Comparison =====
BFS found the shorter path.
BFS Time : 4.4099999286117963e-05
DFS Time : 2.099999983329326e-05
>>>
```

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